

NEW



2026

Ethiopian Health Profession Licensing Examination(EHPLE)

INFORMATION BOOKLET

HUMAN NUTRITION



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MINISTRY OF HEALTH - ETHIOPIA

የዜጎች ጤና ለሃገር ብልጽግና!
HEALTHIER CITIZENS FOR PROSPEROUS NATION!



AAU-Testing Center

Message From the State Minister, Ministry of Health -Ethiopia



H.E Mrs. Frehiwot Abebe

Improving healthcare quality is a global priority for sustainable development, with high quality healthcare being a key component of universal health coverage. One strategy to maintain health care standards is through provision of health professionals' competency assessment. Consequently, in 2019, the Ministry of Health-Ethiopia, initiated the Ethiopian Health Profession Licensing Examination (EHPLE) for undergraduates in seven health disciplines, which has since expanded to include 17 health disciplines.

The main goal of this competency assessment is to identify health professionals with minimal competencies necessary to perform their duties safely and competently, thus enhancing the quality of health care services. This initiative is overseen by a dedicated Health and Health Related Institutions and Professionals' Regulatory Lead Executive Office (LEO), comprising four desks, which plays a pivotal role in strengthening the system and enabling the LEO to conduct the competency exam more extensively and with improved organization and quality.

It is important to note that this competency assessment differs significantly from traditional academic or employment examinations. Hence, this information booklet has been created to address the informational needs of both examinees and teaching faculty regarding the EHPLE. Additionally, it aims to facilitate the assessment process, while promoting transparency and ensuring the sustainability of the program.

The preparation of this booklet involved the collaboration of esteemed experts from various higher education institutions, AAU-IER, the Ministry of Health, JHPIEGO-Ethiopia, Amref, Health Professionals 'Associations, and the Ministry of Education. Their invaluable contributions are acknowledged with sincere gratitude, alongside appreciation for the Ministry of Health staff for their unwavering commitment and hard work throughout the project.

Acknowledgements

This Information Booklet for EHPLE is a contribution from several educators, researchers, students and concerned individuals with a genuine interest to propel Ethiopia's medical and health sciences education forward.

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Acronyms and Abbreviations

EHPLE	Ethiopian Health Profession Licensing Examination
ETA	Educational and Training Authority
HEIs	Higher Education Institutions
HHrIPR-LEO	Health and Health-related Institutions and Professionals Regulatory Lead Executive Office
HSTP-II	Health Sector Transformational Plan-II
MCQ	Multiple Choice Question
MOH	Ministry of Health
WHO	World Health Organization

Purpose of the Information Booklet

The EHPLE Information Booklet serves as a comprehensive guide for those individuals seeking information about the exam. It typically outlines basic information for candidate registration, exam development and administration processes and procedures, result notification, and the licensing process. It also includes information on the exam framework, i.e., the exam domain, sub-domain, content, process, and task, with sample exam items specific to each profession.

The publication of this Booklet is crucial for the following reasons:

- **Clarity and guidance:** It provides clear information about the exam by ensuring candidates understand the necessary information to prepare them.
- **Accessibility:** It serves as a readily accessible resource for individuals pursuing to take the exam, consolidating essential information in one document and facilitating easy access to necessary details. It also helps other stakeholders who might be interested in such resources.
- **Transparency:** It promotes transparency in the examination process and fosters trust among stakeholders about the exam.

In summary, the publication of this Booklet is essential for creating a transparent, standardized, and accessible framework that guides candidates through the EHPLE process.

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Definition of terms

- **Domain:** a broad category or area of knowledge or skills of a profession
- **Sub-domain:** a subset of a broader domain that focuses on knowledge or skills related to the overarching domain
- **Content:** a more specific subcategory, which is a breakdown of the sub-domain
- **Task:** the responsibility, knowledge, skill, and attitude of a junior undergraduate professional in an actual work environment
- **Process:** a systematic sequence of steps or actions designed to achieve a specific outcome
- **Learning outcome:** a clear and measurable statement that describes what the examinee is expected to know or be able to do
- **Relative emphasis:** the proportional importance or weight assigned to different content areas or categories within the assessment
- **Item:** a particular multiple-choice question
- **Item developer:** a subject matter expert responsible for writing test items or questions that make up the examination
- **Item reviewer:** a subject matter expert responsible for reviewing and refining the test items or questions that make up the examination
- **Standard setting:** a process of determining a cut-off point or passing score for an exam
- **Item difficulty index:** a statistical measure that indicates the proportion of examinees who answered a particular test item correctly
- **Discrimination index:** a statistical measure that evaluates how well a particular test item differentiates between high-performing and low-performing examinees
- **Admission paper:** a printout paper generated by the system after completing registration that contains the examinee's photo, QR code, and necessary information

1. Introduction

1.1. Background

Competency assessment is one of the strategies for controlling the standard of healthcare services provided in healthcare facilities. The World Health Organization (WHO) recommends all healthcare professionals to have necessary competencies. In Ethiopia, the Health Sector Transformational Plan-II (HSTP-II) states competency assessment of all graduates before joining the health workforce as one of the strategic initiatives.

The Ministry of Health (MOH) launched the Ethiopian Health Profession Licensing Examination (EHPLE) for undergraduates in 2019. The Health and Health-related Institutions and Professionals Regulatory Lead Executive Office (HHrIPR-LEO) of the Ministry of Health is given a mission to implement the Ministry's objective to achieve competency-related goals. It has the responsibility to ensure that the EHPLE meets technical, professional, and legal standards, and to protect the health, safety, and welfare of the public by assessing candidates' abilities to practice competently.

Currently, the exam is given for 17 health professions (Medicine, Nursing, Public Health, Pharmacy, Medical Laboratory Science, Anesthesia, Midwifery, Dental Medicine, Medical Radiology Technology, Environmental Health, Psychiatric Nursing, Pediatric and Child Health Nursing, Emergency and Critical Care Nursing, Human Nutrition, Surgical Nursing, Optometry, and Physiotherapy). Since its introduction until December 2025, a total of 204,811 examinees took the exam in 23 rounds.

1.2. The Rationale of EHPLE

One of the critical functions of the MoH is to guarantee the efficiency, quality, and equity of healthcare delivery and to protect the public from any undesirable consequences in healthcare delivery practices. As professionals' competence is a significant determinant of the quality of health, evaluation of health professionals' competence has now been given due attention. The licensing examination for health professionals serves as a crucial step to ensure that individuals entering the field meet specific competency standards. The sole aim of the competency assessment is to safeguard public health by verifying that health professionals have the minimal basic knowledge, attitude, and skill required to provide safe and effective care.

Licensing exams act as a preventive measure, ensuring that only competent professionals join the health workforce, which, in turn, contributes to reducing the occurrence of medical errors and enhancing overall patient safety. By setting standards through examinations, regulatory bodies strive to minimize the risk of medical errors caused by incompetence.

2. Key processes of EHPLE

EHPLE involves several key processes to ensure the quality and reliability of the examination.

2.1. Registration of candidates

EHPLE has a mandatory online registration system for both new and repeat candidates, which can be found at <http://hple.moh.gov.et>

Please note these important notes during registration.

New Test Takers:



- The list of eligible candidates from governmental and private Higher Education Institutions (HEIs) will be uploaded to MoH's online registration system by the registrar office of their respective HEIs.
- Candidates who completed their education abroad must bring the following documents in person to MoH in order for their names to be uploaded to the online registration system:
 - ✓ Equivalence document from ETA
 - ✓ Externship attachment completion letter after completing the externship assigned by the regulatory body
- Once the name of the candidate is uploaded to the system and registration has opened for the current exam round, the candidate must register at <http://hple.moh.gov.et> by uploading the necessary documents listed below.
 - ✓ a scanned original or temporary degree
 - ✓ a scanned national ID

Repeat Test Takers:



- Since the information about re-exam candidates already exists in the system, the candidate should register by directly going to <http://hple.moh.gov.et>. There is no need for re-exam candidates to upload their documents.

Both new and repeat candidates:



- After completing the registration, the candidate must download and print the admission paper by logging into his/her account using username and password, or national ID FAN.
- The candidate can change the exam center by logging into his/her account only during the registration period
- An application for center change is not allowed after the registration period is closed.

2.2. Task Analysis

The first step of exam development involves conducting a comprehensive task analysis study, which identifies the tasks, knowledge, skills, and abilities required from a junior undergraduate professional in the specific profession. The analysis is typically done through surveys, interviews, or observations of practitioners in the actual work environment, as well as through the Delphi method with subject matter experts.

2.3. Exam Blueprint

Based on the task analysis findings, the exam blueprint is prepared that outlines the content areas to be covered in the examination and the weight or emphasis given to each area. This ensures that the exam reflects the key competencies and knowledge needed for competent practice in that specific profession. Blueprint is the matrix or chart that shows the number and type of test questions represented across the topics in the content area, consistent with the learning outcome and relative weight of the test given to each content area. The blueprint also identifies the percentage weighting of cognitive dimensions as the level of competence tested in each knowledge domain.

Key components of blueprint are:

- Domain
- Sub-domain
- Content
- Task
- Process
- Learning outcome
- Assessment methods
- Assessment tools/instrument (test format)
- Relative emphasis (in percentage)

2.4. Item Development

The items are developed following specific guidelines to ensure clarity, relevance, and fairness. Subject matter experts with experience in the field are selected from HEIs to develop exam questions (items) that align with the exam blueprint. The exam questions focus mainly on “knows how” according to the competency level of the Miller's pyramid. The items are produced in a secure location on designated computers that are free from internet connectivity. The items are scenario-based and constructed with stem, lead-in, and four options/alternatives. All items will have a single-best-answer type of Multiple-Choice Question (MCQ) that addresses the learning outcome defined in each content area. Standard textbooks, updated guidelines, and standards are used as reference materials.

2.5. Item Review

Once developed, the items undergo a rigorous review process by item reviewers. The main purpose of the exam review process is to evaluate content relevance, technical accuracy, clarity, and sensitivity related to culture and religion. More experienced subject matter experts as well as psychometric experts will do the review to ensure the items meet psychometric standards. Subject matter experts shall review the items to confirm that they are accurate, clearly stated, and correctly keyed using the checklist. Psychometric experts shall review the items to ensure that

they are not technically flawed. They also work on editorial review to check grammar, punctuation, and spelling errors. This helps ensure the reliability and validity of the items.

2.6. Standard setting method

The standard setting or cut-off point of the EHPLE is determined using the Modified-Angoff method, which is one of the most widely used and legally defensible standard setting approaches to set a cut-off point for high-stake competency examinations.

The method involves a panel of subject matter experts who evaluate each exam question and then estimate the probability that a minimally competent examinee would answer each test item correctly. The average of the experts' predictions for an exam question becomes its predicted difficulty. The average of the predicted difficulty values across all items on a test is the recommended cut-off point. This point indicates the minimum level of knowledge and skill required to pass.

2.7. Exam Administration

The EHPLE is administered following established protocols and guidelines. Proper exam administration procedures, appropriate security measures, and appropriate consideration for test-takers who need special support will be applied during exam administration at exam centers. The exam is administered in selected HEIs nationally, where candidates can choose based on their convenience at the time of registration. The exam schedule will be posted ahead of time on the MoH website and social media pages. Examinees who have a valid admission paper are eligible to sit for the exam. The mode of exam administration is computer-based testing.

CAUTIONS

- Candidates are allowed



- Attend the orientation session in order to sit for the exam
- Arrive at the exam center on time
- Bring a legal ID and admission paper
- Complete the exam within the allotted time frame

➤ Candidates are **NOT** allowed



- To bring reference materials, blank paper, or notes into the exam center
- To smoke, eat, or drink in the exam room
- To bring mobile phones, tablets, smart watches, camera devices, eyeglasses, calculators or any type of electronic device into the exam center
- To bring their personal belongings to the exam center
- To bring weapons and sharp materials into the exam center
- To give or receive assistance to or from other candidates during the examination

2.8. Scoring and post exam analysis

Once the exam is completed, the scoring process begins. The exam scoring process involves computerized scoring using software.

Post-exam analysis is the process of analyzing examinees' responses to individual test items in order to assess the quality of the items and the exam as a whole. This phase helps to identify any poorly performing items that may need revision or removal from the exam. The item difficulty index, discrimination index, and reliability coefficient are elements of exam analysis.

2.9. Result notification and appeal management

After scoring and analysis, individual score reports are generated and provided to examinees through the website <http://hple.moh.gov.et>. After result notification, examinees can submit their appeal through phone or email within 10 working days after result notification.

2.10. Licensing

The list of examinees who passed the exam will be sent to regional and city administration regulatory bodies. A license is obtained from the regional/zonal health bodies where he/she permanently lives.

Requirements for professional licensing are:



- Passing the EHPLE
- Original or temporary degree
- Educational documents (10th and 12th certificates)
- Medical certificate
- Government issued ID
- Additional prerequisites based on the requirements of regional regulatory bodies

3. Exam Framework

The key broader professional roles, also known as domains or main knowledge areas, serve as a building framework for the licensing examination content for Human Nutrition professionals. The domains are further divided into discrete professional attributes that constitute sub-units (also referred to as sub-domains) defining the professional identity of Human Nutrition professionals. Tasks specifying the performance level of each sub- domain serve as the final characteristic of the professional duties on which the licensing exam focuses.

The contents of the licensing examination are presented below, structured into key roles (domains), sub-units (sub-domains), and tasks. The examination emphasis for each domain and sub-domain, out of the total 100% questions, is indicated in brackets.

Key professional roles/ domains

- ☐ Nutritional Care (70%)
- ☐ Scholar (7.5%)
- ☐ Professionalism (5.0%)
- ☐ Leadership and management (5.0%)
- ☐ Health promotion and disease prevention (12.5%)

Key role/ domain 1: Patient Care (70%)

Description: This domain encompasses the professional role of Human Nutrition in the provision of high-quality, safety, and patient-centered nutrition service for diverse groups of patients. The provision of up-to-date, ethical and resource-efficient nutrition service required the application of integrated knowledge of clinical, public and behavioral sciences within their scope of practice. As patient care providers, human nutritionists shall possess applied knowledge of anthropometric measurement, health meal habit, SAM and MAM case management and performing appropriate counseling on life stay modification within their scope of practice and with an understanding of the limits of their expertise. To demonstrate competence in this domain, candidates shall apply integrated knowledge in the following sub-units/sub- domains:

- ☐ Clinical nutrition (42%)
- ☐ Public nutrition (28%)

Key role/ domain 2: Scholar (7.5%)

Description: This domain encompasses the professional roles of human nutritionists in generating and utilizing scientific data to improve the health and well-being of the community and broaden their scientific knowledge within the healthcare system and community setting. Providing this service requires the application of integrated knowledge in research methods,

measurements of health and disease, biostatistics, epidemiology, clinical audit, evidence-based practice, and research ethics. To demonstrate competence in this domain, candidates must possess applied knowledge in planning, problem identification, data collection, analysis, interpretation, report writing, and dissemination of research findings.

Key role/ domain 3: Professionalism (5.0%)

Description: This domain encompasses the professional commitment of human nutritionists to promoting the health and well-being of individuals and society through adhering to ethical standards, maintaining personal integrity, and upholding high standards of competence in all areas of practice. To exhibit competence in this domain, candidates must possess applied knowledge of ethical principles, medico-legal practices, effective communication, accountability to the profession and society, maintenance of professional excellence and personal health, continuous professional development and professional values such as, integrity, honesty, altruism, and humility, motivated, compassionate, and respectful care.

Key role/ domain 4: Leadership and management (5.0%)

Description: This domain encompasses the professional roles of human nutritionists in envisioning a high-quality healthcare system through self-awareness, active participation in healthcare team development and leadership, and managing health systems. Providing this service requires the application of integrated knowledge in continuous quality improvement, effective health system leadership, management, and healthcare ethics. To demonstrate competence in this domain, candidates shall possess applied knowledge to plan, organize, staff, lead, execute, monitor, and control healthcare resources and activities.

Key role/ domain 5: Health promotion and disease prevention (12.5%)

Description: This domain encompasses the professional roles of human nutritionists in enhancing the health and well-being of patients, communities, and the larger populations they serve through health advocacy, disease prevention, health promotion, health protection, and the promotion of health equity. Providing this service takes an integrated understanding of determinants of health, health informatics, epidemiology, communicable disease control, and health education.

Table Exam Content for Human Nutrition Profession

Domain_Nutritional Care	
Sub Domain:-Clinical nutrition	
Content	Tasks
Disorders of Nutrient metabolism and Metabolic pathway (Lactose intolerance, Inborn errors of metabolism, Celiac disease)	Assess patients with metabolic disorders
	Diagnose common nutrient metabolism disorders
	Diagnose common fluid & electrolyte imbalances
	Provide dietary management for common metabolic disorders
	Provide dietary management for fluid & electrolyte imbalance
Macronutrient deficiency (Non-edematous, Edematous & marasmic kwashi (mixed), mild-stunting, wasting, underweight)	Assess patients with Protein Energy Malnutrition (PEM)
	Diagnose patients with PEM
	Apply management phases of Severe Acute Malnutrition (SAM) and MAM
	Prevent complication and re-occurrence of SAM & MAM
Micronutrient deficiency (IDA, VAD, IDD, VDD, ZnD, Ca def, VCD, Vit B complex deficiency, Vit K deficiency, Vit E deficiency)	Assess patients with Iron Deficiency Anemia (IDA)
	Assess patients with Vitamin A Deficiency Disorders (VADDs)
	Assess patients with Iodine Deficiency Disorders (IDDs)
	Assess patients with Zinc Deficiency (ZnD)
	Assess patients with other Micronutrient deficiencies (Vitamin D, Calcium, Vitamin C, Vitamin B complex, Vitamin K, Vitamin E)
	Diagnose patients with common Micronutrient deficiencies
	Manage patients with common Micronutrient deficiencies
	Provide counselling on prevention of micronutrient deficiencies
Overnutrition (Overweight, Obesity)	Assess patients with overweight/obesity
	Provide dietary management and lifestyle modifications for Overweight or obesity
Diet related NCDs (DM, CVD, HTN, Cancer, CKD, Gout)	Assess patients with diet related Non-Communicable Diseases (NCDs)
	Diagnose nutritional problems of patients with NCDs
	Apply appropriate intervention to diet related NCDs
Nutrition & Infectious Diseases (IDs) (TB, HIV, Malaria, Intestine parasitosis (IP))	Assess nutritional status of patients with Infectious Diseases (IDs)
	Diagnose nutritional problems in patient with IDs
	Apply appropriate intervention to patient with IDs
Domain_Nutritional Care	
Sub Domain-Public nutrition	
Content	Tasks
Nutrition in Emergency	Assess nutrition status in Emergencies
	Diagnose acute malnutrition during emergency

	Implement Nutritional intervention in emergency
Nutrition through the life cycle (preconception to Geriatrics)	Assess the macro & micronutrient requirements across the human life cycle
	Diagnose common nutritional problems across the human life cycle
	Provide nutritional intervention through the life cycle
Management of Common Nutritional problems of public health importance (PEM, IDD, IDA, VAD, Zn def.)	Apply public health management principles for common nutritional problems in the community
Food & Nutrition Insecurity (Availability, Accessibility, Utilization, Stability)	Assess Food & nutritional insecurity
	Apply possible nutrition intervention mechanisms for food & nutrition insecurity
Domain_Health Promotion & Disease Prevention	
Content	Tasks
Health promotion & education	Apply principles of SBCC (Social, Behavioral Change & Communication)
WaSH & Food hygiene and safety	Apply Water, Sanitation, and Hygiene (WaSH) principles
	Apply Food Safety principles
	Integrate WaSH & Food hygiene and safety principles
	Provide health education on prevention mechanisms of food borne diseases
Domain_Scholar	
Content	Tasks
Research methods in human nutrition, Nutritional epidemiology, Biostatistics	Develop research proposal Apply descriptive statistics analysis
Domain_Professional Ethics	
Content	Tasks
Professional ethics	Adhere to legal and ethical principles
	Demonstrate commitment to Continuous professional Development (CPD)
	Adhere to the food & nutrition policy, programmes, guidelines & standard of practice in Ethiopia
Domain_Leadership and Management	
Content	Tasks
Health Service Management	Apply management function
	Differentiate managerial role and skill
	Identify managerial and leadership style
	Identify principle of resource management
	Differentiate monitoring and evaluation

Sample Questions

1. A 50-year-old man was presented to an OPD with complaints of shortness of breath, chronic chest pain, weakness, and lack of appetite. He also reported palpitation while he used soft drinks or fatty foods. His dietary history revealed with a diet of high fatty food, burger, alcohol and soft drinks regularly. His BMI, MUAC, waist to hip ratio, and skin fold thickness were 29kg/m², 30cm, 0.96, and 0.2mm respectively. His biochemical test showed that HDL Cholesterol level of 30mg/dL. Finally, he was diagnosed with central obesity.

What is the most appropriate anthropometric measurement used to reach the diagnosis in this case?

- A. Body Mass Index
- B. Mid-Upper-Arm Circumference
- C. Waist to Hip Ratio
- D. Skin Fold Thickness

Answer Key: The answer is C

Explanation: Waist-to-hip ratio (WHR) is a direct indicator of central (abdominal) obesity, which reflects fat distribution around the abdomen relative to the hips. In men, a WHR greater than 0.90 and, in women, greater than 0.85 is considered central obesity. Strongly associated with cardiovascular disease, type 2 diabetes, and metabolic syndrome risks (Option C). Body mass index (BMI) measures weight relative to height (weight in kg / height in m²) but do not indicate fat distribution, which is critical for diagnosing central obesity (Option A). MUAC is not used to diagnose central obesity; it assesses general nutritional status, particularly under-nutrition rather than over-nutrition (Option B). Skin Fold Thickness provides an estimate of subcutaneous fat storage, which is useful for assessing energy reserves in malnutrition. Does not measure visceral fat (around internal organs), which is critical in central obesity diagnosis (Option D).

2. A 41-year-old female patient, who was operated on for gastric ulceration four months ago, came to the clinic with a chief complaint of headache and dizziness of four days duration. She had a history of insomnia, heavy duty, and lack of appetite, and she took a large amount of fluid. Her blood pressure was 100/70 mm Hg. Biochemical tests showed a sodium level of 120 mmol/L, a potassium level of 4.0 mmol/L, a calcium level of 3.2 mmol/L, and a hemoglobin level of 12 g/dL.

What is the most likely metabolic problem in this case?

- A. Fluid imbalance
- B. Electrolyte imbalance
- C. Hypertension
- D. Diabetes mellitus

Answer Key: The answer is **B**

Explanation: Both hyponatremia and hypercalcemia are electrolyte disturbances clearly identified in biochemical data. These electrolyte imbalances better explain the patient's neurological symptoms, such as headache and dizziness, than any other option (Option B). Although the patient has a high fluid intake, the main issue indicated by the lab results is low sodium and elevated calcium levels, representing a biochemical imbalance rather than a simple fluid volume problem (Option A). The patient's blood pressure is 100/70 mmHg, which is normal or slightly low, thereby ruling out hypertension (Option C). Furthermore, the patient does not exhibit classic symptoms of diabetes mellitus, such as polyuria, polydipsia, or unexplained weight loss, and the biochemical abnormalities (hyponatremia and hypercalcemia) are not associated with diabetes (Option D).

3. A five-month-old male infant, who showed a decreased breast milk intake for the past one week, was referred to a health center from a health extension worker. His caregiver reported that the child had not been fed well and was not gaining weight. The child appeared to be small for his age, and the nutritionist wanted to determine his nutritional status.

What is the most appropriate anthropometric index measurement to be used in this case?

- A. Weight for height
- B. Height for age
- C. Weight for age
- D. Weight for length

Answer Key: The answer is **D**

Explanation: Weight-for-length is a key anthropometric index used to assess acute malnutrition or wasting in infants and young children. For children under 2 years of age, including infants like this 5-month-old, length is measured lying down (recumbent length) rather than standing height. Weight-for-length is the recommended index to detect wasting in this age group (Option D). "Weight for height" is not used for infants <2 years because they cannot stand for accurate height measurement. In practice, WHO guidelines recommend weight-for-length for infants, not weight-for-height (Option A). Height-for-age will show whether the child has been stunted over a prolonged period but does not assess acute malnutrition or recent weight loss (Option B). Does not differentiate between a child who is wasted (acute malnutrition) and one who is stunted (chronic malnutrition) (Option C).

4. A two-year-old male child is presented with watery diarrhea. His caregiver reported that the child has lost weight. On physical examination the child has bilateral pitting edema on his leg and multiple patchy skins. The health professional suspects edematous malnutrition as a differential diagnosis and wants to assess the nutritional status of the child.

What is the most appropriate anthropometric measurement to be used in this case?

- A. Waist circumference
- B. Mid upper arm circumference
- C. Hip circumference
- D. Head circumference

Answer Key: The answer is **B**

Explanation: MUAC assessing acute malnutrition (wasting) in children aged 6-59 months. Even though the child has edema (kwashiorkor), MUAC is still used to classify severe acute malnutrition (SAM) because weight-based indices (e.g. weight-for-height) are invalidated by edema (Option B). Waist circumference is assessing central obesity and risk of metabolic syndrome, mainly in adults and adolescents and two-year-old with suspected severe acute malnutrition (SAM), waist circumference has no diagnostic value (Option A). Hip circumference is not used in pediatric nutrition assessment to diagnose wasting or edema-related malnutrition (Option C). Head circumference is mainly used in neonatology, growth monitoring, and neurodevelopmental assessments rather than malnutrition screening (Option D).

5. A two-year-old child was brought to a clinic with a complaint of poor growth. The care giver reports that the child has lost his appetite and lost interest in everything. On assessment, the child has a moon face appearance, has multiple patchy skin areas in his abdomen, and some fissures in his groin area.

What is the most important sign needed to reach at a decision of the diagnosis?

- A. Sunken eye
- B. Pale conjunctiva
- C. Bilateral pitting edema
- D. Muscle wasting

Answer Key: The answer is **C**

Explanation: The child shows multiple features (moon face, skin lesions, apathy, poor growth), only bilateral pitting edema confirms kwashiorkor diagnosis. Bilateral pitting edema is diagnostic of kwashiorkor, a type of severe acute malnutrition (SAM) characterized by protein deficiency with adequate or insufficient energy intake (Option C). Sunken eye is a sign of dehydration (often due to diarrhea), not SAM type and relevant in fluid management, not for diagnosing kwashiorkor (Option A). Pale conjunctiva indicates anemia, common in malnourished children but not diagnostic for SAM type (Option B). Muscle wasting suggests marasmus (non-edematous SAM) (Option D).

6. A four-year-old female child was presented to a pediatric clinic with a complaint of low appetite of two weeks duration. Her mother said that she did not gain weight for more than a year. On anthropometric assessment, the z-scores for her weight-for-height, height-for-age, and weight-for-age were -2.0 SD, -2.6 SD, and -1.9 SD, respectively.

What is the most appropriate anthropometric interpretation in this case?

- A. Stunting
- B. Wasting
- C. Normal
- D. Overweight

Answer Key: The answer is A

Explanation: Height-for-age Z-score (HAZ) = -2.6 SD, which is below -2 SD. $HAZ < -2$ SD indicates stunting, meaning chronic malnutrition resulting in impaired linear growth due to prolonged inadequate nutrition or recurrent infections (Option A). This child's HAZ = -2.6 SD confirms stunting. WHZ is -2.0 SD, which is borderline but meets criteria for moderate wasting, however stunting is more severe (HAZ -2.6 SD) and clinically prioritized for chronic nutritional assessment (Option B). None of the indicators are within the normal range (between -2 SD and +2 SD) (Option C). Overweight is indicated by $WHZ > +2$ SD, which is not the case here (WHZ is -2.0 SD) (Option D).

7. A 16-month-old child was brought to a hospital with complaints of intermittent diarrhea, and loss of weight of one week duration. On clinical assessment, the child appeared to be sick but was eager to drink. There was no pertinent finding from urine and stool examination. On anthropometric assessment, MUAC was 12cm, and a z-score for weight-for height was -2.75.

What is the most appropriate interpretation in this case?

- A. Complicated severe acute malnutrition
- B. Uncomplicated severe acute malnutrition
- C. Moderate acute malnutrition
- D. Moderate chronic malnutrition

Answer Key: The answer is C

Explanation: MUAC of 12 cm falls between 11.5 and 12.5 cm, categorizing MAM and WHZ of -2.75 SD is between -2 and -3 SD, also categorizing MAM (Option C). This child has WHZ -2.75 SD, MUAC 12 cm, no edema -does not meet SAM criteria. Additionally, complicated SAM requires: Medical complications and Anorexia (loss of appetite) (Option A). SAM without medical complications or anorexia is called uncomplicated SAM. However, this child does not meet SAM anthropometric criteria, hence cannot be SAM (complicated or uncomplicated) (Option B). The given measurements (MUAC and WHZ) assess acute malnutrition (wasting), not chronic malnutrition (stunting) (Option D).

8. A nine-month-old female child has been presented to a clinic with complaints of watery diarrhea and sunken eyeballs. The mother reports that the child has poor food intake and hasn't been breastfed until the time of examination. On physical examination her MUAC is 11.2 cm, axillary temperature is 36 degrees centigrade, and respiratory rate is 28 breaths per minute. She has no signs of shock, but her skin pinch goes slowly, and she has failed the appetite test.

What is the most appropriate management program option for this child?

- A. Targeted supplementary feeding program
- B. Blanket supplementary feeding program
- C. Outpatient therapeutic program
- D. Stabilization center program

Answer Key: The answer is **D**

Explanation: The child has SAM (MUAC <11.5 cm) and clinical signs of dehydration, plus a failed appetite test and failed appetite test means the child is unable to take therapeutic feeding by mouth, which indicates the need for inpatient care (Option D). Targeted Supplementary Feeding Program (TSFP)-For children with moderate acute malnutrition (MAM), not SAM and child's MUAC indicates SAM (Option A). Blanket Supplementary Feeding Program (BSFP)-community-based prevention for at-risk populations, not for treating SAM or complicated cases (Option B). Outpatient therapeutic program-for uncomplicated SAM cases where the child passes the appetite test and has no medical complications. This child failed an appetite test, so OTP is inappropriate.

9. A three-year-old child was brought to a clinic with poor weight gain. On assessment, her mid-upper arm circumference value is 11.2 cm, her axillary temperature is 36.91 degrees Celsius, her blood sugar level is 51 mg/dl, and her hemoglobin level is 9 g/dl. She had no signs of shock, diarrhea, vomiting, or eye problems. Finally, the child was admitted to the stabilization center.

What is the prior most appropriate treatment for this child?

- A. Amoxicillin for 7 days
- B. 50 ml bolus of 10 percent glucose
- C. 5ml/kg of Resomal
- D. Blood transfusion

Answer Key: The answer is **B**

Explanation: The Blood sugar level = 51 mg/dL, which is hypoglycemia (<54 mg/dL according to WHO SAM guidelines). Hypoglycemia is a life-threatening emergency in severely malnourished children, requiring immediate correction to prevent coma, seizures, or death (Option B). Although empirical antibiotics are part of SAM management, hypoglycemia correction takes priority as it is immediately life-threatening and amoxicillin

would be given after stabilization (Option A). This child has no diarrhea or signs of dehydration, so ReSoMal is not to be an option (Option C). This child's Hb is 9 g/dL, which does not require transfusion (Option D).

10. A one-year-old female child was brought to a health center with a complaint of frequent falling and suspected deformities on her lower extremities. Her mother also reported that her child's knees knock each other whenever she tries to stand up and walk on her own. The child was born during the rainy season. On physical examination, a rachitic rosary was found at the junction of the ribs and cartilage. When the child tried to walk, her legs looked bowed and her knees knocked frequently. Besides, the child had frontal bossing.

What is the most likely nutritional diagnosis in this scenario?

- A. Vitamin D deficiency
- B. Vitamin A deficiency
- C. Vitamin K deficiency
- D. Vitamin E deficiency

Answer Key: The answer is A

Explanation: Vitamin D deficiency is the most likely diagnosis in children with bone deformities, rachitic rosary, frontal bossing, and muscle weakness. It has high clinical relevance due to its impact on growth and lifelong skeletal health (Option A). Vitamin A deficiency does not cause bone deformities, rachitic rosary, or knock knees (Option B). Vitamin K is given at birth to prevent neonatal hemorrhage. Important in liver disease or malabsorption (Option C). Rare except in fat malabsorption disorders or premature infants; supplementation prevents neurological damage (Option D).

11. An environmental health expert of a health center went for monthly supervision to his catchment area. He observed most of the households were using public latrine for excreta disposal. The expert recommended that the community should construct the cheapest type of excreta disposal system which could be constructed using local materials and local skills. It was also considered the most common sanitation system in areas where there was no adequate water supply.

What is the most appropriate type of latrine to be recommended by an expert?

- A. Pit latrine
- B. VIP latrine
- C. Trench latrine
- D. Borehole latrine

Answer Key: The answer is A

Explanation: A Pit latrine is the most appropriate recommendation for rural communities seeking simple, affordable, and sustainable sanitation solutions using local resources (Option A). VIP latrine while better in hygiene and odor control, it is more expensive and requires additional materials and skills compared to a simple pit latrine (Option B). Trench latrine not recommended for long-term community sanitation; lacks privacy, dignity, and sustainability (Option C). Borehole latrine requires specialized equipment and skills to dig narrow deep pits; not cheapest or based on local materials alone (Option D).

12. According to the annual report of a health facility, water- borne diseases were among the top 10 diseases. Based on this information, the health facility planned to conduct a community health education which focused on the prevention and control of water borne diseases.

Which disease prevalence can be decreased after giving the health education in this case?

- A. Trachoma
- B. Malaria
- C. Schistosomiasis
- D. Cholera

Answer Key: The answer is D

Explanation: Health education on water treatment, safe sanitation, and hygiene practices will significantly decrease the prevalence of cholera as it is a classic water-borne disease transmitted via contaminated drinking water (Option D). Trachoma-primarily prevented by improving personal hygiene (face washing) and environmental sanitation, not mainly water-borne ingestion (Option A). Malaria is related to stagnant water as breeding sites, but not a water-borne disease; health education here would focus on vector control, not safe water use (Option B). Schistosomiasis is water-contact disease, but not water-borne through drinking; prevention focuses on avoiding infested water and snail control, not water treatment for drinking (Option C).

13. A nutritionist working in a primary hospital approached his manager to talk about his frustration with the work assignment in the hospital. The nutritionist believes that he is under-serving the hospital because of his current ward assignment. He feels overburdened by the amount of task assigned to him and by the fact that he is not using his specialization which puts him under lots of pressure to carry out his task.

What is the most appropriate organizing action addressing the above concern?

- A. Decentralize nutritionist task to others
- B. Review scheme of division of labor
- C. Delegate some of the duties to others
- D. Coordination of activities in the ward

Answer Key: The answer is **B**

Explanation: Reviewing the scheme of division of labor is the most appropriate managerial action to ensure tasks are appropriately assigned according to staff specialization and capacity, preventing overload and enhancing job satisfaction (Option B). May lead to tasks being assigned to unqualified personnel, compromising service quality. Does not address the issue of appropriate utilization of his specialization (Option A). He may not have the authority to delegate, especially if tasks require his specialization. The problem lies in managerial planning of task allocation, not personal delegation (Option C). Important for efficiency but does not address task overload or specialization mismatch experienced by the nutritionist (Option D).

14. A clinical nutritionist challenged by a patient having a poor progress of anemia after treated by hem-up for the last two weeks. Then, the clinician decided to search the best available evidence online and got many primary articles done on it which made him to confuse for utilization. Finally, with his team decided to use best article over others based on standard. What is the most likely study design to utilize evidence in the above scenario?
- A. Cross-sectional
 - B. Case-Control
 - C. Experimental
 - D. Cohort

Answer Key: The answer is **C**

Explanation: For evaluating treatment efficacy (like Hem-up for anemia), experimental studies, especially randomized controlled trials, provide the highest level of primary evidence to guide clinical decision-making effectively (Option C). Useful for prevalence studies, not for establishing treatment efficacy (Option A). Useful for identifying risk factors, not for assessing effectiveness of an intervention (Option B). Provides strongest evidence for treatment efficacy (Option D).

15. A nutrition health professional working in a primary hospital was striving to be committed to work in the hospital and update his capacity to practice safely, effectively and legally within a scope of practice. Based on self-initiated plans he had participated in three formal trainings organized on newly adopted guidelines on nutritionist related professional practices. What is the most likely health care practicing the nutritionist demonstrated in this scenario?
- A. Professionalism
 - B. Development in Ethics
 - C. Improving in career structure
 - D. Continuing professional development

Answer Key: The answer is **D**

Explanation: Continuing Professional Development (Option D) involves actively updating and expanding professional knowledge and skills through formal trainings, workshops, self-learning, and reflective practices, ensuring safe, effective, and legal practice within one's professional scope (Option D). Although participation in CPD demonstrates professionalism, the scenario specifically focuses on learning and updating skills, which is CPD (Option A). The trainings were about new clinical guidelines, not ethical frameworks specifically (Option B). the scenario describes skills and knowledge development, not structural career movement (Option C).



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